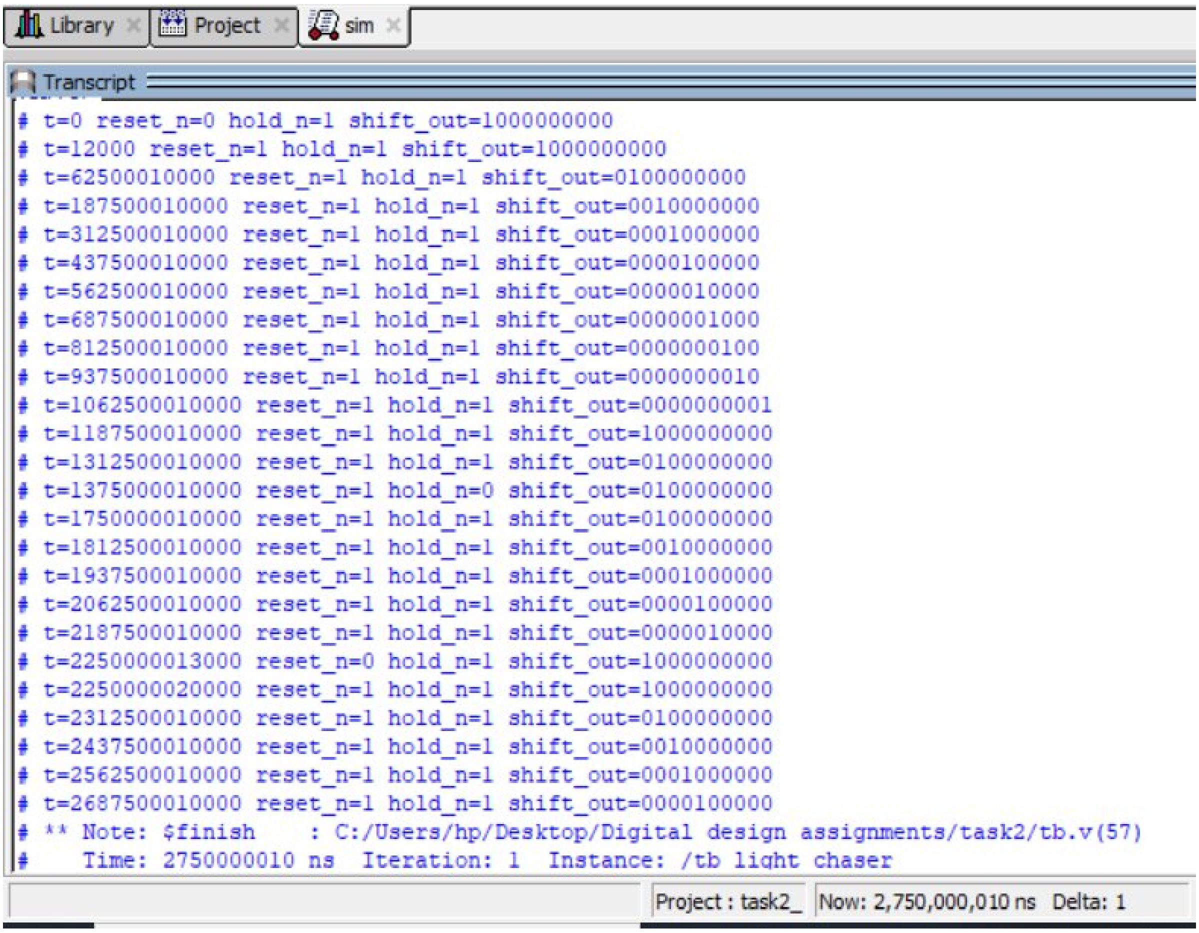
Light chaser

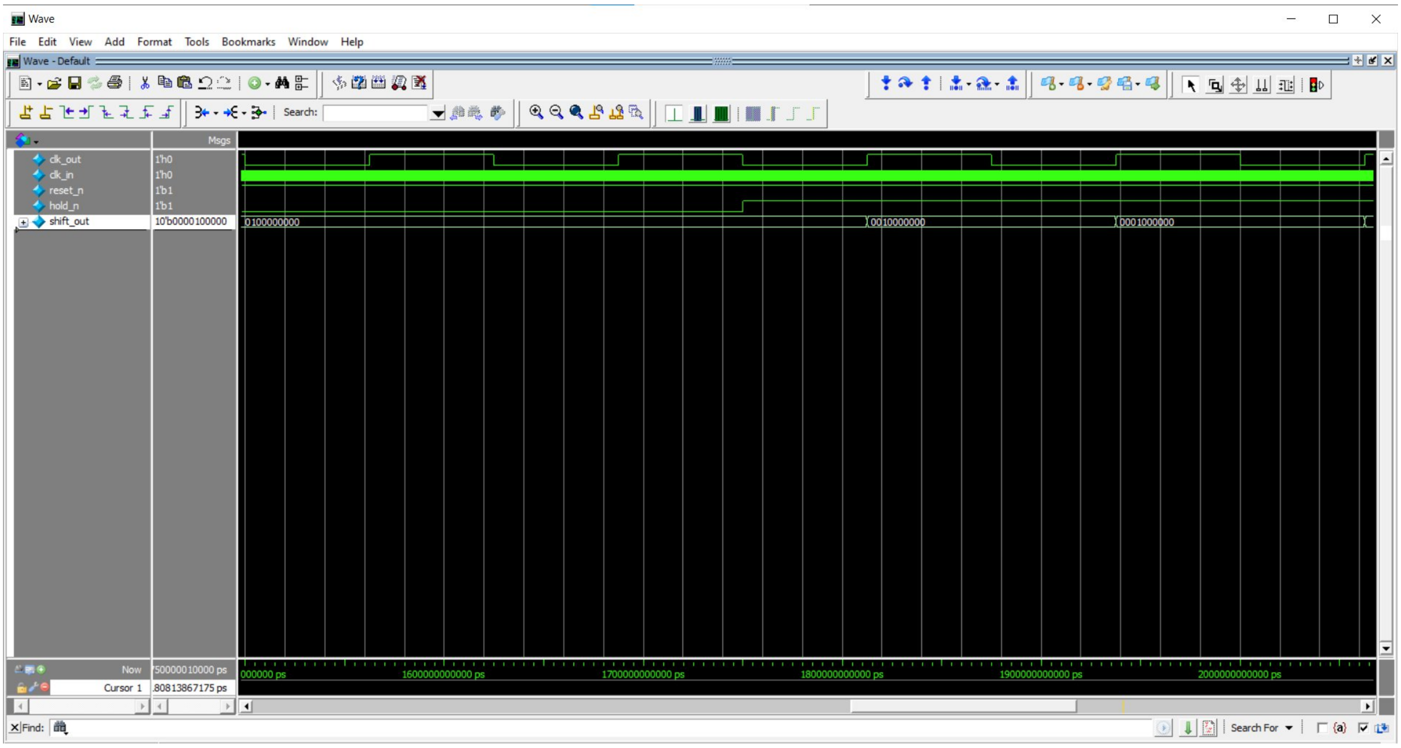
Test cases:

Test case	Stimulus	Expected behavior	Result
Free-run shift	reset_n de-asserted, hold_n=1	One-hot bit shifts once per clk_out period, full 10-bit wraparound	Pass
Hold	hold_n=0 for 3 clk_out periods	shift_out remains constant	Pass
Resume after hold	hold_n returns to 1	Shifting resumes from last state	Pass
Async reset mid-shift	reset_n=0 pulsed mid-sequence	shift_out immediately returns to initial state, independent of clock edge	Pass
Clock division accuracy	Measured clk_out period vs. calculated	125 ms measured vs. 125 ms calculated (50 MHz/6.25M divisor)	Match

Where, 187,500,010,000 − 62,500,010,000=125,000,000,000 ns
125,000,000,000 ns ÷ 1,000,000=125,000 ms
DIVISOR × T_clk_in period = 6,250,000×20 ns=125,000,000 ns=125 ms

Result:
All test cases produced outputs matching the expected values:




Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
Search:
Msgs
clk_out 1'h0
clk_in 1'h0
reset_n 1'b1
hold_n 1'b1
shift_out 10'b0000100000
0100000000
0010000000
0001000000
Now
Cursor 1
000000 ps
1600000000000 ps
1700000000000 ps
1800000000000 ps
1900000000000 ps
2000000000000 ps
Find:
Search For